Band-1 Computational Experiment: Hadamard Matrices - Single 4k Orders

Human + ZoraASI full-dash (executor)

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Band-1 Computational Experiment: Hadamard Matrices - Single Orders of Form $4k$

Honesty stamp. This note does **not** claim that a Hadamard matrix exists for every multiple of 4. The **full Hadamard conjecture is Band-3 / OPEN**. Tonight is **single-order Band-1**: a literature ledger of historically contested and currently reported-open orders, plus on-box constructions (Sylvester, Paley) and an exhaustive normalized search for one small known order, each certified by the exact identity $HH^{\mathrm{T}} = nI$.

Abstract

A Hadamard matrix of order n is an $n \times n$ matrix with entries ± 1 whose rows are pairwise orthogonal, equivalently $HH^{\mathrm{T}} = nI$. Necessary: $n \in \{1, 2\}$ or $4 \mid n$. We (i) ledger the twelve multiples of $4 \leq 2000$ that were open as of ~2014, the August 2026 announcement filling those gaps, and the smallest commonly cited remaining order **2004**; (ii) implement Sylvester doubling and Paley types I/II over prime fields, plus Kronecker composition; (iii) run a normalized backtracking search that **finds** Hadamard matrices of orders **4, 8, and 12** from scratch. All constructions and search hits are verified in exact integer arithmetic. **Full conjecture status: OPEN.** Order **2004** was **not** searched (intractable tonight).

Problem statement (single-order Band-1)

Existence question for a fixed admissible n: does there exist $H \in \{\pm 1\}^{n \times n}$ with $HH^{\mathrm{T}} = nI$?

- Band-1 (tonight): settle or certificate-check **individual** small / historically contested orders; build constructive toolkit.
- Band-3: prove existence for **all** $n = 4k$.

Literature ledger (not re-proved on box)

Era / item	Status / orders	Citation / note
Necessary condition	$n = 1, 2$ or $4 \mid n$	Classical
Sylvester 1867	all 2^k	Kronecker / doubling
Hadamard 1893	12, 20	Explicit examples
Paley 1933	$q + 1$ ($q \equiv 3 \pmod 4$ prime power); $2(q + 1)$ ($q \equiv 1 \pmod 4$)	Quadratic character
Baumert-Golomb-Hall 1962	**92** (first not Sylvester+Paley)	Williamson computer search
Kharaghani-Tayfeh-Rezaie 2005	**428**	*J. Combin. Designs*
Djoković 2007	**764**	arXiv:math/0703312

Era / item	Status / orders	Citation / note
~2014 ledger	**12** open orders ≤ 2000: 668, 716, 892, 1132, 1244, 1388, 1436, 1676, 1772, 1916, 1948, 1964	OEIS A007299 / Wikipedia
Aug 2026 Alpöge-Voinov-Reynolds-Haertle (+ Claude)	Announced explicit matrices for **all 12**; independent checks reported (VibeMathed, Nechita)	X announcement; Wikipedia update; ion.nechita.net
Sep 2026 survey	Smallest commonly cited remaining gap: **2004**	Engineered.at / VibeMathed / news summaries
Full conjecture	**OPEN** (Band-3)	-

Contest / uncertainty. Pre-2026, the smallest contested order was **668**. After the August 2026 announcement and reported exact verifications, orders ≤ 2000 are treated here as **reported-settled** (checkable certificates; archival journal citation may lag). **2004** remains the smallest order with **no** verified construction in surveyed sources. We do **not** re-decode or re-verify the Aug 2026 payloads on box tonight.

Method (box)

1. **Certificate.** Given H, check entries $\in \{\pm 1\}$ and $\max |HH^{\mathrm{T}} - nI| = 0$ in `int64` arithmetic.
2. **Sylvester.** $H_1 = [1]$; $H_{2m} = \begin{bmatrix} H & H \\ H & -H \end{bmatrix}$. Orders $1, 2, 4, \ldots, 128$.
3. **Paley I.** For prime $q \equiv 3 \pmod 4$: core $Q_{ij} = \left(\frac{j-i}{q}\right)$; $H = \begin{bmatrix} 1 & \mathbf{1}^{\mathrm{T}} \\ \mathbf{1} & Q - I \end{bmatrix}$ (order $q+1$).
4. **Paley II.** For prime $q \equiv 1 \pmod 4$: conference matrix $C = \begin{bmatrix} 0 & \mathbf{1}^{\mathrm{T}} \\ \mathbf{1} & Q \end{bmatrix}$ ($CC^{\mathrm{T}} = qI$); $H = \begin{bmatrix} C+I & C-I \\ C-I & -(C+I) \end{bmatrix}$ (order $2(q+1)$).
5. **Kronecker.** If H_a, H_b are Hadamard then so is $H_a \otimes H_b$ (order ab).
6. **Exhaustive search (normalized).** First row and first column fixed to $+1$; backtrack remaining entries with partial row-orthogonality pruning. Applied to $n = 4, 8, 12$.

Not attempted: exhaustive / heuristic search for order **2004** (or 668 payloads). Order **52** is admissible ≤ 64 but not reached by prime-field Paley + Sylvester + Kronecker alone tonight (Paley II over GF(5^2) would give 52; prime-power fields deferred).

Results (numbers!)

Unit tests

`PASS`: H_1, H_2, Sylvester H_4, Paley I order 12 ($q = 11$); all-ones 4×4 correctly **rejected**.

Constructions verified ($HH^{\mathrm{T}} = nI$ exactly)

32 distinct orders certified on box, including:

Family	Orders (sample)	Failures
Sylvester 2^k	1, 2, 4, 8, 16, 32, 64, 128	**0**
Paley I ($q \equiv 3 \pmod 4$)	4, 8, 12, 20, 24, 32, 44, 48, 60, 68, 72, 80, 84	**0**
Paley II ($q \equiv 1 \pmod 4$)	12, 28, 36, 60, 76, 84, 108, 124	**0**
Kronecker composites	24, 40, 48, 56, 80, 88, 96, 112, 144, 160, 176, 224, ...	**0**

Full verified list: **1, 2, 4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 56, 60, 64, 68, 72, 76, 80, 84, 88, 96, 108, 112, 124, 128, 144, 160, 176, 224**.

Coverage of admissible orders ≤ 64: all except **52** (see method note). Wall-clock for full probe ~= **0.13 s**.

Exhaustive normalized search (methods validation)

n	Found?	Nodes	Time (s)	Verified $HH^{\mathrm{T}} = nI$?
4	yes	17	0.00015	yes
8	yes	529	0.0017	yes
12	yes	14384	0.046	yes

Order **12** is the classical first non-pure-Sylvester order; the search recovers an explicit matrix without invoking the Paley formula, then the same certificate checks it.

Open / contested ledger (honesty)

- Historically open ≤ 2000 (2014 list of 12): **reported filled** Aug 2026 (Alpöge et al.); not re-verified on box tonight.
- **Smallest reported open order now: 2004 - no** construction attempted; exhaustive search **not tractable** for Band-1 tonight.
- **Full Hadamard conjecture: OPEN (Band-3).**

Reproducibility

- Code: `/workspace/experiments/band1/hadamard/`
- Results: `results.json`
- Literature: `literature_ledger.json`
- Log: `/workspace/experiments/logs/2026-09-11_hadamard.log`
- Run: `python3 hadamard_check.py`

Conclusion

Band-1 deliverable: constructive certificates for **32** Hadamard orders (Sylvester / Paley / Kronecker) plus exhaustive-search hits for **4, 8, 12**, all with $\max |HH^{\mathrm{T}} - nI| = 0$. **Literature:** pre-2026 gap list of twelve orders ≤ 2000 reported closed in August 2026; smallest cited remaining single order **2004** left untouched as intractable tonight. **Does not** settle the full conjecture.